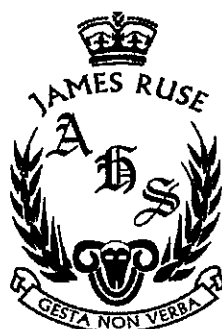


Student name _____

Class _____



YEAR 12

ASSESSMENT TEST 1
TERM 4, 2018

MATHEMATICS EXTENSION 1

Time Allowed – 90 Minutes

(Plus 5 minutes Reading Time)

- All questions may be attempted
- NESA approved calculators and templates are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged work
- NESA Reference sheets are provided

The answers to all questions are to be returned in separate bundles clearly labeled Question 1, Question 2, etc.

Each question must show (in the top right hand corner) your Candidate Number.

Question 1 (9 marks)

- (a) Find $\int \frac{1}{\sqrt{4-x^2}} dx$ 1
- (b) Find the exact value of $\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{2dx}{1+4x^2}$ 2
- (c) Find $\int x\sqrt{3x+2} dx$ using the substitution $u = 3x + 2$. 3
- (d) Find the value of the constant term in the expansion of $\left(3x + \frac{2}{\sqrt{x}}\right)^6$. 3

Question 2 (9 marks) [START A NEW PAGE]

- (a) For the function $y = \pi + 2 \tan^{-1}(3x)$
- (i) State the range of the function. 1
- (ii) Find the value of y when $x = \frac{1}{3}$ 1
- (iii) Sketch the curve of the function showing all information from previous parts. 2
- (b) Find the volume when the area between the curves $y = x^3$ and $y = 4x$, in the first quadrant is revolved about y - axis. The two curves intersect at points (0,0) and (2,8). 3
- (c) When $(5 + 3x)^n$ is written out as a polynomial in x , the coefficients of x^5 and x^6 have the same value. Find n . 2

Question 3 (9 marks) [START A NEW PAGE]

- (a) Find the exact value of:
- (i) $\cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)$ 1
- (i) $\sin\left[\tan^{-1}\left(-\frac{1}{2}\right)\right]$ 2
- (b) Determine the co-ordinates of the point where the curve $y = 3e^{\sin^{-1}x}$ cuts the y axis. Find the gradient of the tangent to the curve at this point. 3
- (c) Determine the coefficient of a^3 in the expansion of $(1 - 3a)^2(1 + 2a)^5$ 3

Question 4 (9 marks) [START A NEW PAGE]

(a) Let $f(x) = x^2 - 6x + 2$.

- (i) By expressing $f(x)$ in the form $(x - h)^2 + k$, or otherwise, find the range of $f(x)$. 1
- (ii) The domain of $f(x)$ is restricted to the largest domain such that it has an inverse. The domain contains $x = 5$. Find the equation of the inverse function $f^{-1}(x)$. 2
- (iii) Find the point(s) of intersection of $y = f(x)$ and $y = f^{-1}(x)$, given that $f(x)$ maintains its restricted domain. 2

- (b) (i) T_k is the k^{th} term in the expression $\left(\frac{1}{3} + 2x\right)^{18}$. Find $\frac{T_{k+1}}{T_k}$ and simplify. 2
- (ii) Hence, find the greatest co-efficient in the expansion $\left(\frac{1}{3} + 2x\right)^{18}$. 2

Question 5 (9 marks) [START A NEW PAGE]

- (a) A region R is bounded by the curve $y = 2\sin^{-1} x$, the x -axis and the line $x = 1$. Show this information on a diagram and use Simpson's rule with five function values to give an approximation for the area R . Give your answer correct to two decimal places. 4

(b) (i) Prove that $\frac{d}{dx} \left[\log_e x - \frac{1}{2} \log_e(1 + x^2) - \frac{\tan^{-1} x}{x} \right] = \frac{\tan^{-1} x}{x^2}$. 3

(ii) Hence, find $\int_2^3 \frac{\tan^{-1} x}{x^2} dx$

Give your answer to two decimal places. 2

Question 6 (9 marks) [START A NEW PAGE]

(a) (i) Write down the general solution for $\sin^2 \theta = \frac{3}{4}$. 2

(b) Use the substitution $x = u - 2$ to evaluate $\int_0^1 \frac{2 - x}{(2 + x)^3} dx$. 4

(c) Find the derivative of $\sin^{-1} \sqrt{1 - x}$. Simplify your answer. 3

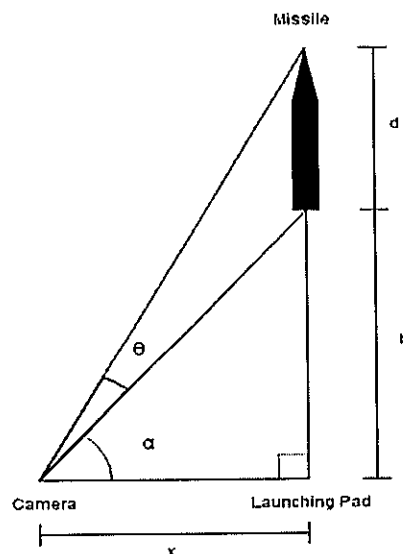
Question 7 (9 marks) [START A NEW PAGE]

(a) (i) Given that $\cos \theta = \frac{1}{\sqrt{2}} \sin \alpha + \frac{1}{2\sqrt{2}} \cos \alpha$, find an expression for θ . **1**

(ii) Find the derivative $\frac{d\theta}{d\alpha}$ and solve $\frac{d\theta}{d\alpha} = 0$. Leave your answer in the exact form. **2**

(b) A camera is located x metres from the base of a missile launching pad.

Let α be the angle of elevation of the base of the missile from the camera lens.



(i) If a missile of length d metres is launched vertically upwards, show that when the base of the missile is h metres above the camera lens level, the angle θ subtended at the lens by the missile is given by:

$$\theta = \tan^{-1} \left(\frac{h+d}{x} \right) - \tan^{-1} \left(\frac{h}{x} \right). \quad \text{1}$$

(ii) You are given that $h = 60 \text{ m}$ and $d = 20 \text{ m}$. Show that θ is maximised when the camera is $40\sqrt{3}$ metres from the launching pad. **3**

(iii) Deduce that the maximum angle subtended by the missile at the camera lens is $\tan^{-1} \left(\frac{1}{4\sqrt{3}} \right)$ **2**

END OF EXAMINATION

MATHEMATICS: Question

Suggested Solutions

Marks

Marker's Comments

1. a) $\sin^{-1} \frac{x}{2} + C$

①

b) $\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{2}{1+4x^2} dx$

$= \frac{1}{4} \int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{2}{\frac{1}{4}+x^2} dx$

$= \int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{\frac{1}{2}}{\frac{1}{4}+x^2} dx$

$= \left[\tan^{-1}(2x) \right]_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}}$

$= \tan^{-1} \sqrt{3} - \tan^{-1} 1$

$= \frac{\pi}{3} - \frac{\pi}{4}$

$= \frac{\pi}{12}$

① for recognising $\tan^{-1}(\text{no.})$

but NO marks if $\tan^{-1}(x^2)$!

c) $I = \int x \sqrt{3x+2} dx$

let $u = 3x+2$

$\frac{du}{dx} = 3$

$dx = \frac{du}{3}$

$= \frac{1}{9} \int \sqrt{u} (u-2) du$

$= \frac{1}{9} \int u^{\frac{3}{2}} - 2u^{\frac{1}{2}} du$

$x = \frac{u-2}{3}$

$= \frac{1}{9} \left[\frac{2u^{\frac{5}{2}}}{5} - \frac{4u^{\frac{3}{2}}}{3} \right] + C$

$= \frac{1}{9} \left[\frac{2(3x+2)^{\frac{5}{2}}}{5} - \frac{4(3x+2)^{\frac{3}{2}}}{3} \right] + C$

①

①

①

MATHEMATICS: Question

Suggested Solutions

Marks

Marker's Comments

$$d) \left(3x + \frac{2}{\sqrt{x}}\right)^6$$

$$= \sum_{r=0}^6 {}^6C_r (3x)^{6-r} \left(2x^{\frac{1}{2}}\right)^r$$

$$\text{constant} \rightarrow x^{6-r} x^{-\frac{r}{2}} = x^0$$

$$x^{6-\frac{3r}{2}} = x^0$$

$$\therefore 6 - \frac{3r}{2} = 0$$

$$\frac{3r}{2} = 6$$

$$3r = 12$$

$$r = 4$$

$$\therefore \text{constant} = {}^6C_4 3^2 2^4$$

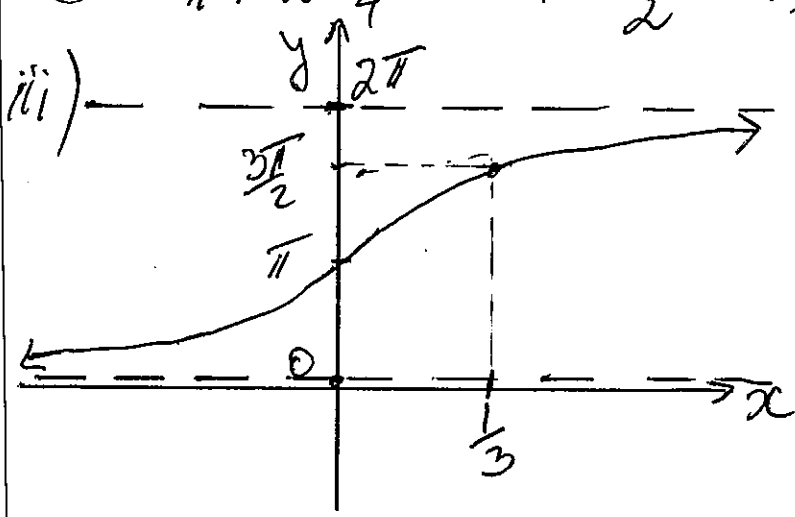
$$= 2160$$

Careful $\left(3x + \frac{2}{\sqrt{x}}\right)^6 \neq {}^6C_r (3x)^{6-r} (2x^{\frac{1}{2}})^r$

You need a summation sign or
say "general term equals"!

!!!

MATHEMATICS Extension 1 : Question... 2

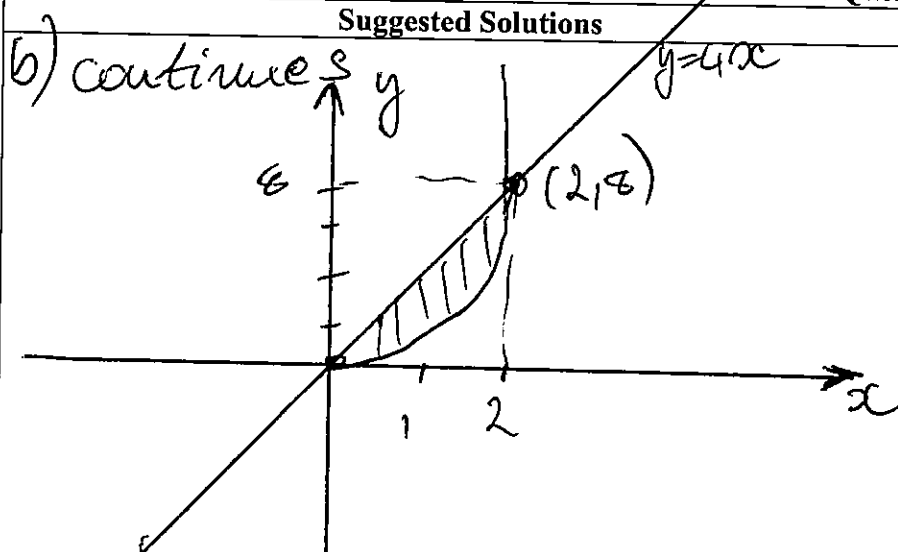
Suggested Solutions	Marks	Marker's Comments
a) i) $\pi + 2 \tan^{-1}(3x)$ $-\frac{\pi}{2} < \tan^{-1} 3x < \frac{\pi}{2}$ $-\pi < 2 \tan^{-1} 3x < \pi$ $0 < \pi + 2 \tan^{-1} 3x < 2\pi$ $\therefore$ Range is $0 < y < 2\pi$ ii) $y = \pi + 2 \tan^{-1}(3 \times \frac{1}{3})$ $= \pi + 2 \frac{\pi}{4} = \pi + \frac{\pi}{2} = \frac{3\pi}{2}$ iii) 	① ① ①	* done well * if $\leq$ given, is fixed on graph * if made mistake here, CTE for the graph * another easy mark * for full 2 marks: * correct asympt. * correct points of intersection * correct slope
b) $y = x^3$ 1st quadrant $y = 4x$ about y-axis $y^{\frac{1}{3}} = x$ $y^{\frac{1}{4}} = x$ $V = \pi \int_a^b x^2 dy$	①	

MATHEMATICS Extension 1 : Question.....2

Suggested Solutions

Marks

Marker's Comments



$$V = \pi \int_0^8 \left(y^{1/3}\right)^2 dx - \pi \int_0^8 \left(\frac{y}{4}\right)^2 dy$$

$$= \pi \int_0^8 \left(y^{2/3} - \frac{y^2}{16}\right) dy$$

$$= \pi \left[\frac{3y^{5/3}}{5} - \frac{y^3}{48} \right]_0^8$$

$$= \pi \left[\frac{3}{5} 8^{5/3} - \frac{8^3}{48} - 0 \right] =$$

$$= \frac{128}{15} \pi u^3$$

4

* i / done
(over x axis
only 1 mark
(awarded
as: you make
it easier &
did not read
question
& correctly

①

* number of
you find the
 $\left(y^{1/3} - \frac{y}{4}\right)^2$ which
is incorrect,
just imagine
your 3D shape

①

MATHEMATICS Extension 1 : Question...2...

Suggested Solutions

Marks

Marker's Comments

$$c) (5+3x)^n = \sum_{r=0}^n 5^{n-r} (3x)^r \binom{n}{r}$$

$$T_7 = \binom{n}{6} 5^{n-6} 3^6 x^6$$

$$T_6 = \binom{n}{5} 5^{n-5} 3 x^5$$

equating

$$\binom{n}{6} 5^{n-6} 3^6 = \binom{n}{5} 5^{n-5} 3$$

$$\frac{n!}{(n-6)! 6!} \times \frac{5^4 3^6}{5^6} = \frac{n!}{(n-5)! 5!} \times \frac{5^4 3^5}{5^5}$$

(3)

$$\frac{(5 \times (n-6) \times (n-7) \dots 1 \times (6 \times 5 \times \dots \times 1))}{1} =$$

$$= \frac{((n-5) \times (n-6) \times (n-7) \dots 4 \times 5 \times 4 \times \dots \times 1)}{1}$$

$$\frac{3}{6} = \frac{5}{n-5}$$

$$3n - 15 = 30$$

$$3n = 45 \Rightarrow n = 15$$

(1)

(1)

* number of
you have

n-5=10
and got
n=5!!!???

JRAHS : TASK 1 : 3U : 2018 T4

MATHEMATICS Extension 1: Question 3

Suggested Solutions

Marks
Awarded

Marker's Comments

a) (i) $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = x$

$\therefore \cos x = -\frac{1}{\sqrt{2}}$

$\Rightarrow x = \frac{3\pi}{4} \text{ or } \frac{5\pi}{4}$

But $0 \leq \cos^{-1} m \leq \pi$

$\therefore \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) = \frac{3\pi}{4} \text{ only}$

1

For Correct,
in radians

(ii) Method I:

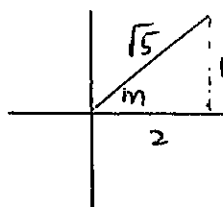
$\sin\left(\tan^{-1}\left(-\frac{1}{2}\right)\right)$

$= \sin\left(-\tan^{-1}\frac{1}{2}\right)$ as $\tan^{-1}(-x) = -\tan^{-1}x$

$= -\sin\left(\tan^{-1}\frac{1}{2}\right)$ as $\sin^{-1}(-x) = -\sin^{-1}x$

Let $m = \tan^{-1}\frac{1}{2}$

$\Rightarrow \tan m = \frac{1}{2}$

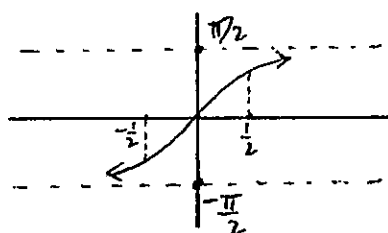


$\Rightarrow \sin m = \frac{1}{\sqrt{5}}$

$-\frac{\pi}{2} < \tan^{-1}x < \frac{\pi}{2}$

(1) $\rightarrow$ restriction must be stated, OR
 $0 < \tan^{-1}\frac{1}{2} < \frac{\pi}{2}$

$\therefore \sin\left(\tan^{-1}\left(-\frac{1}{2}\right)\right) = -\frac{1}{\sqrt{5}}$



(1) $\rightarrow$ sign and correct exact value

JRAHS TASK 1 : 3u : TERM 4 : 2018

MATHEMATICS Extension 1: Question 3

Suggested Solutions

Marks
Awarded

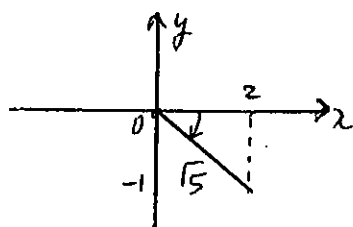
Marker's Comments

Method II:

$$\text{Let } \tan^{-1}\left(-\frac{1}{2}\right) = x$$

$$\therefore \tan x = -\frac{1}{2}$$

$$\text{But } -\frac{\pi}{2} < \tan^{-1}\left(-\frac{1}{2}\right) < 0$$



$$\therefore \sin x = -\frac{1}{\sqrt{5}}$$

$$\text{i.e. } \sin\left(\tan^{-1}\left(-\frac{1}{2}\right)\right) = -\frac{1}{\sqrt{5}}$$

1

For correct
restriction

1

For correct
exact value

$$\text{b) } y = 3e^{\sin^{-1}x}$$

$$\begin{aligned} \text{When } x=0, y &= 3e^{\sin^{-1}0} \\ &= 3e^0 \\ &= 3 \end{aligned}$$

$\therefore$ curve cuts the y-axis at (0, 3)

$$y' = \frac{3}{\sqrt{1-x^2}} e^{\sin^{-1}x}$$

$$\text{When } x=0, y' = \frac{3}{\sqrt{1-0}} \cdot e^0 = 3$$

$$\therefore \text{gradient} = 3$$

1

Coordinates of
y-intercept
For derivative

1

1

For gradient

MATHEMATICS Extension 1: Question 3		
Suggested Solutions	Marks Awarded	Marker's Comments
c) <u>Method I</u> : $(1-3a)^2(1+2a)^5$ $= (1-6a+9a^2)(1+5(2a)+10(2a)^2+10(2a)^3+\dots)$ Terms in a^3 are : $1 \times 80 a^3$, $-6 \times 10 \times 4 a^3$ $9 \times 10 a^3$. $\therefore$ Coefficient of a^3 is $80 - 240 + 90 = -70$ <u>Method II</u> : $(1-3a)^2(1+2a)^5 = (1-6a+9a^2) \sum_{k=0}^5 {}^5C_k (2a)^k$ $= \sum_{k=0}^5 {}^5C_k 2^k (a^k - 6a^{k+1} + 9a^{k+2})$ For $k=1$, Coefficient of a^3 is ${}^5C_1 \cdot 2^1 \cdot 9 = 90$ $k=2$, " " " " ${}^5C_2 \cdot 2^2 \cdot (-6) = -240$ — 1 $k=3$, " " " " ${}^5C_3 \cdot 2^3 \cdot 1 = 80$ $\therefore$ Coefficient of a^3 in $(1-3a)^2(1+2a)^5$ is -70 — 1	1 1 1	For applying Pascal's Δ to expand using Pascal's Δ For correct Terms For correct Coefficient

MATHEMATICS Extension 1: Question 3

Suggested Solutions

Marks
Awarded

Marker's Comments

Method III :

$$(1-3a)^2 (1+2a)^5$$

$$= \sum_{r=0}^2 {}^2C_r (-3a)^r \sum_{k=0}^5 {}^5C_k (2a)^k$$

$$= \sum_{r=0}^2 \sum_{k=0}^5 {}^2C_r {}^5C_k (-3)^r a^r \cdot 2^k \cdot a^k$$

$$= \sum_{r=0}^2 \sum_{k=0}^5 {}^2C_r {}^5C_k (-3)^r (2)^k a^{r+k}$$

To obtain coefficient of a^3 ,
set $r+k=3$

$$r=0, k=3 \Rightarrow \text{Coefficient is } {}^2C_0 {}^5C_3 (-3)^0 (2)^3 = 80$$

$$r=1, k=2 \Rightarrow \quad \quad \quad " \quad \quad {}^2C_1 {}^5C_2 (-3)^1 (2)^2 = -240$$

$$r=2, k=1 \Rightarrow \quad \quad \quad " \quad \quad {}^2C_2 {}^5C_1 (-3)^2 (2)^1 = 90$$

$\therefore$ Net coefficient is -70

1

For applying
Binomial theorem

1

For correct
Coefficient

MATHEMATICS Extension 1: Question 4		
Suggested Solutions	Marks	Marker's Comments
(a) (i) $f(x) = (x - 3)^2 - 7$ implies $f(x) \geq -7$ for all real x. So, the range is $\{y \in \mathbf{R} \mid y \geq -7\}$.	1	No significant problems.
(ii) By inspection of the quadratic, f is increasing for all $x \in \mathbf{R}$ such that $x \geq 3$ and this is the domain containing 5. Inverse: let $y = f^{-1}(x)$. Then $x = f(y)$ $= (y - 3)^2 - 7$ $y = 3 \pm \sqrt{x + 7}$ Now, range of f^{-1} is the domain of f $\text{rng } f^{-1} = [3, \infty) \subset \mathbf{R}$ which means we take the positive solution. So, $f^{-1}(x) = 3 + \sqrt{x + 7}$	1	Again, handled well. Candidates reasoned appropriately to restrict domain and determine appropriate branch of $\pm\sqrt{}$ to use.
(iii) The graphs of f and f^{-1} intersect on $y = x$. So, solve either $f(x) = x$ or $f^{-1}(x) = x$. Here, $x^2 - 6x + 2 = x$ $x^2 - 7x + 2 = 0$ So, $x = \frac{7 \pm \sqrt{41}}{2}$ And since $\text{rng } f^{-1} = \text{dom } f = \{x \in \mathbf{R} \mid x \geq 3\}$, must take	1	Two major issues: (1) candidates not realising that intersection of f and f^{-1} occurred on the line $y = x$, and so attempted to solve $f = f^{-1}$ simultaneously, leading to a quartic that was intractable given the time; (2)

$x = \frac{7 + \sqrt{41}}{2}$ Since $y = x$ at intersection, the point is $\left(\frac{7 + \sqrt{41}}{2}, \frac{7 + \sqrt{41}}{2} \right)$ (b) Let T_k be the k^{th} term in the expansion of $\left(\frac{1}{3} + 2x \right)^{18}$ Note that $k = 1, 2, 3, \dots$ (i) $T_k = \binom{18}{k-1} \left(\frac{1}{3} \right)^{18-(k-1)} (2x)^{k-1}$ So, $\frac{T_{k+1}}{T_k} = \frac{\binom{18}{k} \left(\frac{1}{3} \right)^{18-k} (2x)^k}{\binom{18}{k-1} \left(\frac{1}{3} \right)^{18-(k-1)} (2x)^{k-1}}$ $= \left[\frac{18!}{k! (18-k)!} 3^{k-18} 2^k x^k \right] \div \left[\frac{18!}{(k-1)! (18-k+1)!} 3^{k-19} 2^{k-1} x^{k-1} \right]$ $= \frac{6(19-k)}{k} x$ (ii) To find the greatest coefficient, we want the greatest integer k such that $\frac{6(19-k)}{k} \geq 1$ $114 - 6k \geq k$ $7k \leq 114$ $k \leq 16 \frac{2}{7}$ So, $k = 16$ here. The greatest coefficient is therefore $\binom{18}{16} \left(\frac{1}{3} \right)^{18-16} 2^{16} = 1\,114\,112$	1 candidates realising that intersection occurred on $y = x$, solving correctly for x, the substituting the value back into the function to derive y...giving either a mess or an approximation of 6.7. Only significant problem with this situation was not realising that terms begin their count at $k = 1$, so the k^{th} term would correspond to the coefficient aligned with $\binom{18}{k-1}$. 1 Candidates weren't penalised for this if their subsequent reasoning was correct. 1 Important to realise that the 'term' includes variables (some students left out x). 1 Great majority of candidates were able to derive the correct coefficient.
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Q5

Ext 1 2018 T4

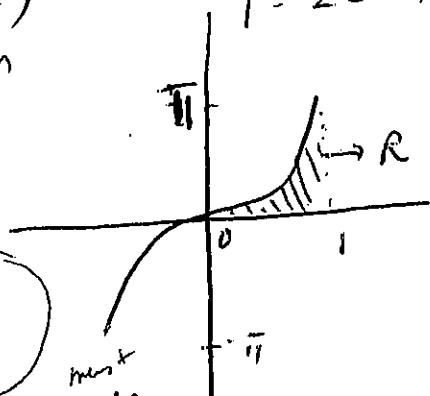
P1/2

a)
4m

(1)

must
shade
the region

$$y = 2 \sin^{-1} x$$



x	0	0.25	0.5	0.75	1
y	0		$\frac{\pi}{3}$		$\frac{\pi}{2}$

1.0472

3.1416

$$x=0.5 \quad y = 2 \frac{\pi}{6} = \frac{\pi}{3} = 1.0472$$

0.5054

$$x=0.25 \quad y = 2 \sin^{-1} \frac{1}{4} =$$

$$x=0.75 \quad y = 2 \sin^{-1} \frac{3}{4} = 1.6961$$

if they
did not
mark π
on y-axis
no penalty.

x	0	0.25	0.5	0.75	1
y	0	0.5054	1.0472	1.6961	3.1416

(1m)

$$\frac{1}{12} \left[1 \times f(0) + 4 f\left(\frac{1}{4}\right) + 2 f\left(\frac{1}{2}\right) + 4 f\left(\frac{3}{4}\right) + f(1) \right] \quad (1m)$$

$$A = \frac{0.5}{6} \left[4 \left[2 \sin^{-1} \frac{1}{4} + 2 \sin^{-1} \frac{3}{4} \right] + 2 \times \frac{\pi}{3} + \pi \right]$$

$$= \frac{1}{12} \left[8.806 + 2.0944 + 3.1416 \right]$$

$$= \frac{14.042}{12}$$

use Radian

$$= \underline{\underline{1.170}} \text{ m}^2 \quad (1)$$

$$= \underline{\underline{1.17}} \text{ m}^2 \text{ (2dp)}$$

x	y	(x)	(y)
0	0	1	0
$\frac{1}{4}$	$2 \sin^{-1} \left(\frac{1}{4} \right)$	4	$8 \sin^{-1} \left(\frac{1}{4} \right)$
$\frac{1}{2}$	$\frac{\pi}{3}$	2	$2 \frac{\pi}{3}$
$\frac{3}{4}$	$2 \sin^{-1} \left(\frac{3}{4} \right)$	4	$8 \sin^{-1} \left(\frac{3}{4} \right)$
		1	π

4 x 0.5054

2 x 1.0472

4 x 1.6961

3.1416

b i) $y = \ln x - \frac{1}{2} \ln(1+x^2) - \frac{\tan^{-1} x}{x}$ $-\left[\frac{\frac{2x}{1+x^2} - \tan^{-1} x}{x^2} \right]$

3m

$$y' = \frac{1}{x} - \frac{\frac{1}{2} \cdot \frac{2x}{1+x^2}}{1} + \left(\frac{\tan^{-1} x}{x^2} + \left(\frac{1}{x} \cdot \frac{1}{1+x^2} \right) \right)$$

must get + line to 2m

$$y' = \frac{1}{x} - \frac{x}{1+x^2} + \frac{\tan^{-1} x}{x^2} + \frac{1}{x(1+x^2)}$$

$$y' = \frac{1+x^2-x^2}{x(1+x^2)} + \frac{\tan^{-1} x}{x^2} + \frac{-1}{x(1+x^2)}$$

★ be careful of the signs in

$$y' = \frac{\tan^{-1} x}{x^2}$$

$$-\frac{\tan^{-1} x}{x}$$

ii) $\int_2^3 \frac{\tan^{-1} x}{x^2} dx = \left[\ln x - \frac{1}{2} \ln(1+x^2) - \frac{\tan^{-1} x}{x} \right]_2^3$

$$= \left(\ln 3 - \frac{1}{2} \ln 10 - \frac{\tan^{-1} 3}{3} \right) - \left(\ln 2 - \frac{1}{2} \ln 5 - \frac{\tan^{-1} 2}{2} \right)$$

$$= \left(\ln 3 - \ln \sqrt{10} - \frac{\tan^{-1} 3}{3} \right) - \left(\ln 2 - \ln \sqrt{5} - \frac{\tan^{-1} 2}{2} \right)$$

$$= \ln \frac{3}{2} - \ln \sqrt{10} + \ln \sqrt{5} + \frac{\tan^{-1} 2}{2} - \frac{\tan^{-1} 3}{3}$$

$$= \ln \left(\frac{3}{2} \frac{\sqrt{5}}{\sqrt{10}} \right) + \frac{\tan^{-1} 2}{2} - \frac{\tan^{-1} 3}{3}$$

$$= \ln \left(\frac{3}{2\sqrt{2}} \right) + \frac{1}{2} \tan^{-1} 2 - \frac{\tan^{-1} 3}{3}$$

$$= 0.05889 + 0.5536 - 0.4163$$

$$= 0.19619 \approx 0.1962 \text{ (2dp)}$$

MATHEMATICS Extension 1 : Question...6...

Suggested Solutions

Marks

Marker's Comments

$$a) \sin^2 \theta = \frac{3}{4}$$

$$\sin \theta = \pm \frac{\sqrt{3}}{2}$$

$$\therefore \theta = n\pi + (-1)^n \left(\pm \frac{\pi}{3} \right), n \in \mathbb{Z}$$

$$= n\pi \pm \frac{\pi}{3}$$

1 $\Rightarrow$ 1 solution
2 $\Rightarrow$ 2 solution plus
 $n \in \mathbb{Z}$

$$b) \int_0^1 \frac{2-x}{(2+x)^3} dx$$

$$\text{let } x = u - 2$$

$$\frac{dx}{du} = 1$$

$$\therefore dx = du$$

$$\text{when } x=1, u=3$$

$$x=0, u=2$$

$$\therefore \int_0^1 \frac{2-x}{(2+x)^3} dx = \int_2^3 \frac{2-(u-2)}{(2+u-2)^3} du$$

$$= \int_2^3 \frac{4-u}{u^3} du$$

$$= \int_2^3 \left(\frac{4}{u^3} - \frac{1}{u^2} \right) du$$

$$= \left[-2u^{-2} + u^{-1} \right]_2^3$$

$$= \left(-\frac{2}{9} + \frac{1}{3} \right) - \left(-\frac{1}{2} + \frac{1}{2} \right)$$

$$= \frac{1}{9}$$

1 $\Rightarrow$ upper and lower
limits

1 correct integral
in terms of u

$$c) \frac{d(\sin^{-1} \sqrt{1-x})}{dx} = \frac{-\frac{1}{2}(1-x)^{-\frac{1}{2}}}{\sqrt{1-(\sqrt{1-x})^2}}$$

$$= \frac{-1}{2\sqrt{1-x}}$$

$$= \frac{-1}{2\sqrt{1-x}}$$

$$= \frac{-1}{2\sqrt{x}\sqrt{1-x}}$$

$$= \frac{-1}{2\sqrt{x}\sqrt{1-x}}$$

1

1

1

MATHEMATICS Extension 1 : Question...7....

Suggested Solutions

Marks

Marker's Comments

a) (i) $\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}} \sin \alpha + \frac{1}{2\sqrt{2}} \cos \alpha\right)$

1

(ii) $\frac{d\theta}{d\alpha} = \frac{-\left(\frac{1}{\sqrt{2}} \cos \alpha - \frac{1}{2\sqrt{2}} \sin \alpha\right)}{\sqrt{1 - \left(\frac{1}{\sqrt{2}} \sin \alpha + \frac{1}{2\sqrt{2}} \cos \alpha\right)^2}}$

1

derivative

$= 0$

$\therefore \frac{1}{\sqrt{2}} \cos \alpha = \frac{1}{2\sqrt{2}} \sin \alpha$

$\tan \alpha = 2$

$\alpha = \tan^{-1} 2$

1

value of α

b) (i) $\tan \alpha = \frac{h}{x}$, $\tan(\alpha + \theta) = \frac{h+d}{x}$

1

$\alpha = \tan^{-1} \frac{h}{x}$, $\alpha + \theta = \tan^{-1} \frac{h+d}{x}$

$\therefore \theta = \tan^{-1} \frac{h+d}{x} - \tan^{-1} \frac{h}{x}$

(ii) $\theta = \tan^{-1} \frac{80}{x} - \tan^{-1} \frac{60}{x}$

$\frac{d\theta}{dx} = \frac{-\frac{80}{x^2}}{1 + \left(\frac{80}{x}\right)^2} - \frac{-\frac{60}{x^2}}{1 + \left(\frac{60}{x}\right)^2}$

$= \frac{-80}{x^2 + 80^2} + \frac{60}{x^2 + 60^2}$

1

derivative

For maximum $\frac{d\theta}{dx} = 0$

$\frac{60}{x^2 + 60^2} = \frac{80}{x^2 + 80^2}$

MATHEMATICS Extension 1 : Question...7....

Suggested Solutions

Marks

Marker's Comments

$$3x^2 + 3 \times 80^2 = 4x^2 + 4 \times 60^2$$

$$x^2 = 4800$$

$$x = \sqrt{4800}, x > 0$$

$$= 40\sqrt{3}$$

x	60	$40\sqrt{3}$	80
$\frac{d\theta}{dx}$	0.0025	0	0.0025

/ — \

$\therefore$ Since the function is continuous and there is only one stationary point for $x > 0$ then $x = 40\sqrt{3}$ gives the maximum value of θ .

(iii) When $x = 40\sqrt{3}$

$$\theta = \tan^{-1} \frac{80}{x} - \tan^{-1} \frac{60}{x}$$

$$= \tan^{-1} \frac{2}{\sqrt{3}} - \tan^{-1} \frac{3}{2\sqrt{3}}$$

$$\tan \theta = \tan(\tan^{-1} \frac{2}{\sqrt{3}} - \tan^{-1} \frac{3}{2\sqrt{3}})$$

$$= \frac{\tan(\tan^{-1} \frac{2}{\sqrt{3}}) - \tan(\tan^{-1} \frac{3}{2\sqrt{3}})}{1 + \tan(\tan^{-1} \frac{2}{\sqrt{3}}) \tan(\tan^{-1} \frac{3}{2\sqrt{3}})}$$

$$= \frac{\frac{2}{\sqrt{3}} - \frac{3}{2\sqrt{3}}}{1 + \frac{2}{\sqrt{3}} \times \frac{3}{2\sqrt{3}}}$$

$$= \frac{\frac{2}{\sqrt{3}} - \frac{3}{2\sqrt{3}}}{1 + \frac{2}{\sqrt{3}} \times \frac{3}{2\sqrt{3}}}$$

$$= \frac{\frac{2}{\sqrt{3}} - \frac{3}{2\sqrt{3}}}{1 + \frac{2}{\sqrt{3}} \times \frac{3}{2\sqrt{3}}}$$

MATHEMATICS Extension 1 : Question...7...

Suggested Solutions

Marks

Marker's Comments

$$= \frac{4-3}{2\sqrt{3}} \div (1+1)$$

$$= \frac{1}{2\sqrt{3}} \div 2$$

$$= \frac{1}{4\sqrt{3}}$$

$$\therefore \theta = \tan^{-1} \frac{1}{4\sqrt{3}}$$

1